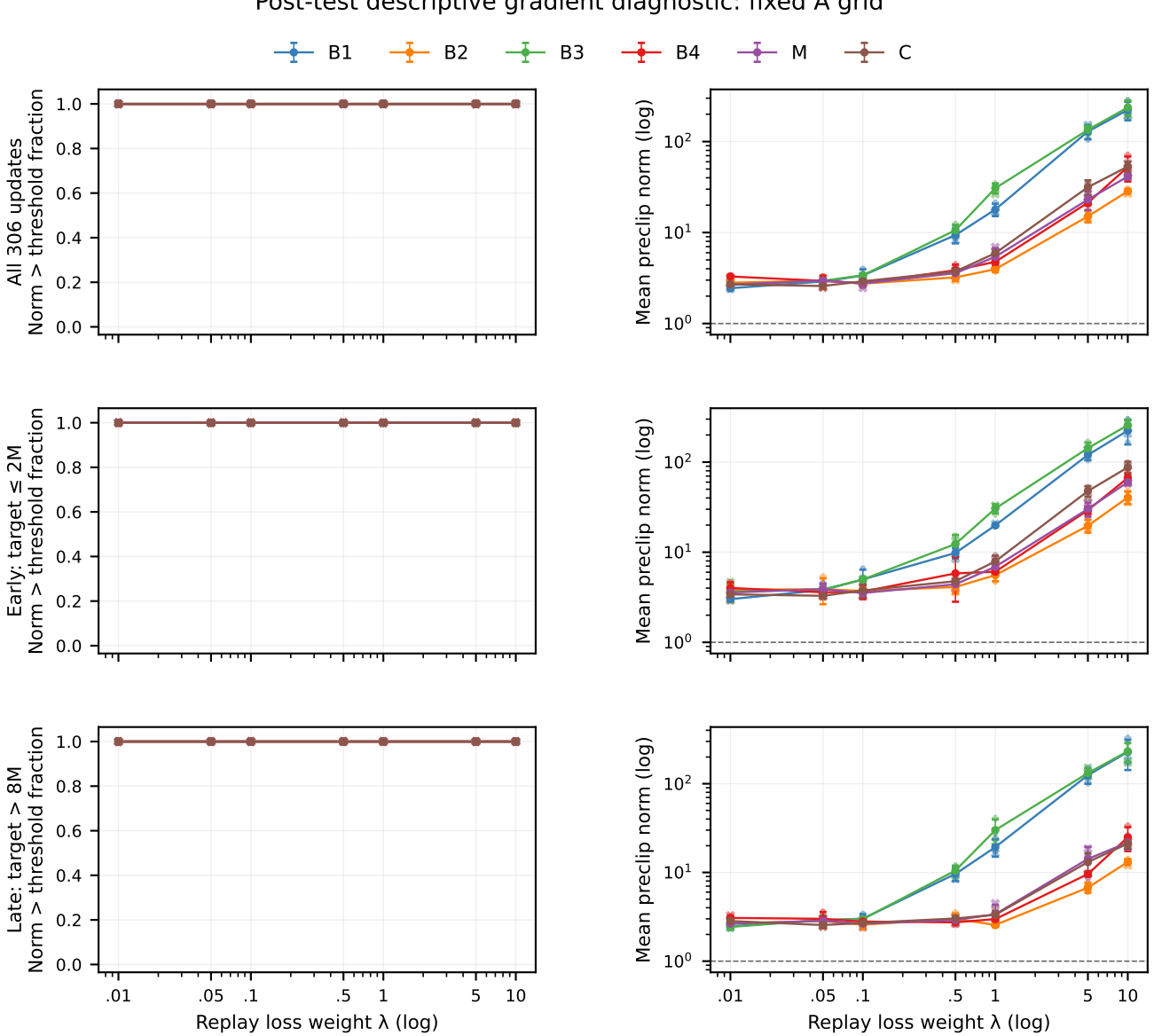


All three pipeline seeds; mean ± sample SD. Steps/windows and λ cells share random sources.
Threshold exceedance is not an Adam-update or causal-mechanism measure.